

1.4 Cognitive psychology — memory and forgetting

Students will be assessed on their ability to:

- 1 **Definition of the approach**
 - a Define cognitive psychology showing understanding that the approach is about the role of cognition/cognitive processes in human behaviour.
 - b Define and use psychological terminology accurately and appropriately including the terms:
 - i information processing
 - ii memory
 - iii forgetting
 - iv storage
 - v retrieval.

2 Methodology/ How Science Works

- a Identify, describe and apply the following terms:
 - i natural, laboratory and field experiment
 - ii independent variable (IV) and dependent variable (DV)
 - iii experimental hypothesis
 - iv directional (one tailed) and non-directional (two tailed)
 - v repeated measures, matched pairs and independent groups design
 - vi operationalisation of variables
 - vii counterbalancing
 - viii randomisation
 - ix order effects.
- b Describe and evaluate, including strengths and weaknesses, the experimental method (laboratory, natural, field) in terms of:
 - i experimental control (including the effects of situational and participant variables)
 - ii objectivity
 - iii reliability
 - iv validity
 - v experimenter effects
 - vi demand characteristics.

3 Content

- a Describe and evaluate the Levels of Processing framework for memory research (Craik and Lockhart, 1972) and **one other** theory or model of memory (other than cue dependent theory).

Suitable examples:

- reconstructive memory (Bartlett, 1932)
- multi-store model of memory (Atkinson & Shiffrin, 1968)
- working memory (Baddeley & Hitch, 1974)
- spreading-activation Model of Semantic Memory (Collins & Loftus, 1975).

- b Describe and evaluate the Cue Dependent Theory of Forgetting (Tulving, 1974) and **one other** theory of forgetting.

Suitable examples:

- displacement
- trace decay
- interference theory.

4 Studies in detail

- a Describe and evaluate in detail Godden and Baddeley's (1975) study of cue dependent forgetting/memory and **one other** study of memory or forgetting in the Cognitive Approach. This **must** be selected from the following:

- Peterson & Peterson (1959) Suppression of rehearsal and the role of interference
- Craik and Tulving (1975) Levels of processing
- Ramponi et al (2004) Levels of processing effects on involuntary or implicit recall and age.

5 Key issue

- a Describe **one** key issue of relevance to today's society and apply concepts, theories and/or research (as appropriate) drawn from the Cognitive Approach to explain the issue.

Suitable examples:

- flashbulb memory
- reliability of eyewitness testimony
- the use of the cognitive interview.

Note: in examination, students may be given stimulus material from a key issue to explain using concepts, theories and/or research (as appropriate) from the Cognitive Approach.

6 Evidence of practice: short experiment

- a Devise and conduct **one** practical, which must be an experiment, to gather data relevant to a topic covered in the Cognitive Approach for this course. This experiment must be designed and conducted according to ethical principles.
- Suitable examples:
- interference task on short-term memory
 - levels of processing task
 - state or context dependency forgetting task.
- b Comment on the research design decisions.
- c Collect, present and comment on data gathered including using measures of central tendency (mean, median, mode), measures of dispersion (at least range), bar graph, histogram, frequency graph as relevant.

Methods

Note: students must be prepared to answer exam questions focused on practical work, which will include questions about the practical exercises themselves and questions about the general methodological issues that are specified for this particular unit. This will include the following requirement.

- d Identify, describe and apply the following:
 - i natural, laboratory and field experiment
 - ii independent and dependent variables
 - iii experimental (directional and non-directional) hypothesis
 - iv repeated measures, matched pairs and independent groups designs
 - v control over participant and situational variables
 - vi measures of central tendency
 - vii measures of dispersion (at least range)
 - viii bar graph, histogram and frequency graph as ways to present data collected
 - ix experimenter effects and demand characteristics
 - x objectivity, validity, reliability
 - xi operationalisation
 - xii counterbalancing, randomisation, order effects.